

Highlights

The continuum limit of a space-filling layer of cubes: a tiling identity, an exact lattice kernel and the closed-form error of the discrete layer

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- Voxel coupling averaged over the source cell equals the continuum's exactly
- The cube's single-site T-matrix follows from distributional moment integrals
- The voxel scheme's leading error is known in closed form for every grid
- A first-moment voxel converges at fourth order for P, SV and SH waves
- In a random stratified layer, one voxel per model layer reaches 1e-6 accuracy

The continuum limit of a space-filling layer of cubes: a tiling identity, an exact lattice kernel and the closed-form error of the discrete layer

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Abstract

We analyse the convergence of voxel discretisations of the elastic volume integral equation, in which a medium is divided into cubic cells, each represented by a single-site T -matrix and coupled through the background Green's tensor: the elastic counterpart of the discrete dipole approximation. Because cubes tile space, a layer of identical voxels is exactly a homogeneous layer, so it isolates the discretisation error, with no modelling or shape error. We show that (i) the cube's lateral form factor vanishes on the reciprocal lattice, so the source-cell-averaged coupling equals the continuum's; (ii) the single-site T -matrix follows from distributional moments of the Green's tensor and, against exact scattering by a sphere, is exact statically and departs only by a real, closed-form $(ka)^2$ term; (iii) its self term cancels, so the scheme is a collocation of the continuum equation, of second order with leading error $+(kd)^2/8$ in reflection and $-(kd)^2/24$ in transmission, which no scalar correction can remove; (iv) a Galerkin voxel carrying Legendre moments of its internal field to degree p converges at order $2p + 2$, with leading error $-(kd)^4/720$ at $p = 1$, for incident P, SV and SH waves at normal and oblique

incidence, and one plane of it reproduces the thin-layer series in the thickness D through $O(D^{2p+2})$. In a randomly stratified layer the orders are unchanged, and the fourth-order voxel reaches an accuracy of 10^{-6} with one voxel per model layer. On a voxelised sphere the staircase error dominates. Every result is verified by an independent implementation or an exact solution.

Keywords: Elastic wave scattering, Volume integral equation, Foldy–Lax multiple scattering, Discrete dipole approximation, Galerkin method, Convergence analysis

1. Introduction

The problem this paper serves. Seismic waves in the crust are scattered by heterogeneity on every scale. The models that describe it are parameterised in many ways (as layers or blocks, by smooth basis functions, or by interfaces), and a calculation that uses one must first sample it onto a grid. A volume-integral scattering calculation represents the sampled model as cells of constant properties and treats every cell as a scatterer: a voxel carrying a single-site T -matrix and coupled to every other voxel through the Green’s tensor of the background, with the multiple-scattering series summed. For the media such a scheme is built for, there is no exact solution, and that is the difficulty. Its error cannot be measured against the answer. Comparison with another numerical method mixes the errors of both. And a refinement study can mislead, because a voxel scheme can converge smoothly to the wrong limit. §4 shows how that happens with the most natural choice of coupling, the Green’s tensor evaluated between voxel centres: it carries a local term whose size per unit volume does not shrink as the voxels do, and unless the

single voxel's response is matched to cancel it, refinement converges to the wrong answer without revealing it.

Why a homogeneous layer. This paper therefore starts from the simplest configuration in which a voxel scheme runs in full (every voxel, every coupling, the whole multiple-scattering series, at normal and at oblique incidence) and the exact answer is nevertheless known: a layer filled with identical voxels. Because cubes tile space, such a layer does not approximate a homogeneous layer; it *is* one. Any difference between its reflection and transmission and those of the continuous layer is therefore error of the discretisation alone, with no modelling error and no shape error behind which to hide. A sphere, the usual test object for volume-integral and discrete-dipole methods [1], cannot give this: its error mixes the discretisation with the staircase approximation of a curved surface by cubes. The layer plays the part of the patch test in finite elements: reproducing a uniform medium correctly is necessary, though not sufficient, for a scheme to be trusted in a heterogeneous one. What the layer establishes about the scheme's two ingredients, the coupling between voxels and the response of a single voxel, are properties of the ingredients, which are the same in any medium: that the averaged coupling is exact, that the self term cancels, and the form of the single-site error. §9 confirms that they carry over to a stratified medium; a laterally heterogeneous one remains to be tested.

What the reader gains. Four things that a heterogeneous test cannot provide. First, a proof, exact at normal incidence, that the voxel scheme converges to the continuum, and the specific ingredient, a coupling averaged over the source voxel, without which it converges to the wrong answer. Second, the

42 leading error in closed form for every discretisation, so that a grid can be
 43 chosen for a required accuracy before the calculation is run. Third, a single-
 44 site T -matrix derived from first principles, with its error stated and checked
 45 against the exact solution for a sphere. Fourth, a voxel that converges at
 46 fourth order, and with further Legendre moments at any even order, also in
 47 a randomly stratified model (§9): for a reflection accuracy of 10^{-6} it allows
 48 voxels 58 times larger in each direction than the standard one (§11).

49 *Why cubic voxels.* Cubes are the natural cell for three reasons. They *tile*
 50 *space*, so a uniform region discretised into cubes is still that uniform region,
 51 whereas spheres or point scatterers can only approximate it and carry a pack-
 52 ing error of their own. Their *regular lattice* makes the interaction between
 53 cells translation invariant, so the coupling is a convolution (Toeplitz between
 54 planes, Bloch-periodic within them), and the whole system is solved by fast
 55 transforms. And a cube small compared with the wavelength has a *single-site*
 56 *response in closed form*, the elastic depolarisation of a cube (§6), so the only
 57 approximation a voxel introduces is the internal field it assumes.

58 *Multiple scattering on a lattice.* Replacing a heterogeneous body by a lattice
 59 of small scatterers that interact through the background Green's function
 60 is the multiple-scattering description of Foldy [2] and Lax [3]. In electro-
 61 magnetics its discretised form is the discrete dipole approximation [DDA;
 62 4, 5], whose accuracy has been studied in detail: each dipole's polarisability
 63 must account for the local field of the surrounding lattice, which the lattice
 64 dispersion relation fixes so that an infinite lattice propagates waves as the
 65 continuum does to a given order in kd [6]; the error of the method converges
 66 at second order in the dipole spacing when shape errors are absent [1]; and

67 integrating the Green’s tensor over each cell rather than sampling it at the
68 centre improves accuracy, most of all for strong contrasts [7]. The elastic
69 problem adds a structure the scalar and electromagnetic ones lack. A cubic
70 voxel responds to the incident field with a force (through its density contrast)
71 and a moment (through its stiffness contrast), so the state at a voxel has nine
72 components (displacement and strain), and the single site’s self-interaction is
73 the elastic depolarisation of a cube, the discrete counterpart of the inclusion
74 tensor of Eshelby [8].

75 *This paper.* The approach originates in the two-and-a-half-dimensional for-
76 mulation of Nestor [9]: a medium divided into cells that interact through the
77 background Green’s tensor, each carrying a single-site contrast renormalised
78 by its own near field, with the multiple-scattering series summed. This pa-
79 per extends it to cubic voxels. §2 sets out the discrete medium and §3 the
80 continuous layer it must reproduce. §4 shows why the point coupling fails,
81 and §5 why the source-averaged coupling succeeds exactly. §6 derives the
82 single-site T -matrix from moment integrals, tests it against the exact sphere,
83 and shows that with it the discrete system is a collocation of the continuum
84 equation. §7 gives that collocation’s second-order error in closed form, and
85 §8 the fourth-order voxel, for incident P and S at normal and oblique inci-
86 dence, and its generalisation to Legendre moments of any degree. §9 repeats
87 the tests on a randomly stratified layer, §10 on a sphere, and §11 discusses
88 what the results imply for voxel schemes in general. Every result is derived
89 symbolically in Mathematica and checked against an independent Python
90 implementation or an exact solution.

91 2. Setting

92 *The discrete medium.* Space is tiled by cubes of side d (half-side $h = d/2$).
 93 Cube i carries a contrast $\Delta\lambda, \Delta\mu, \Delta\rho$ from an isotropic background with
 94 Lamé parameters λ, μ , density ρ , P and S speeds α, β and P-wave modulus
 95 $M_P = \lambda + 2\mu$. Time dependence is $e^{-i\omega t}$; $k_P = \omega/\alpha$, $k_S = \omega/\beta$. Coordinates
 96 are ordered (z, x, y) with z down. The field at a voxel is the nine-component
 97 state

$$\psi = (u_z, u_x, u_y, \varepsilon_{zz}, \varepsilon_{xx}, \varepsilon_{yy}, 2\varepsilon_{xy}, 2\varepsilon_{zy}, 2\varepsilon_{zx}), \quad (1)$$

98 displacement followed by engineering strain. A voxel responds with a force
 99 and a moment (stress dipole); the 9×9 single-site T -matrix maps the exciting
 100 state to them, and the 9×9 lattice Green's tensor $\mathbf{K}(i, j)$ maps a source
 101 voxel's force and moment to the state at voxel i . The Foldy–Lax system is

$$\psi_i = \psi_i^0 + \sum_{j \neq i} \mathbf{K}(i, j) \mathbf{T}_j \psi_j, \quad (2)$$

102 with ψ^0 the incident state. Strain rows of $\mathbf{K}$ are receiver derivatives of the
 103 point Green's tensor $\mathbf{G}$; moment columns are derivatives with respect to the
 104 receiver coordinate, symmetrised with a factor $\frac{1}{2}$ on the shear components.
 105 In that convention the contrast operator is

$$\Delta = \text{blockdiag}(\omega^2 \Delta\rho \mathbf{I}_3, \Delta\mathcal{C}), \quad (3)$$

106 where $\Delta\mathcal{C}$ is the 6×6 contrast stiffness with $\Delta\lambda + 2\Delta\mu$ and $\Delta\lambda$ in the normal
 107 block and $2\Delta\mu$ (not $\Delta\mu$) on the shear diagonal.

108 *The laterally uniform layer.* A layer of thickness D is filled with n planes
 109 of voxels, $d = D/n$, and illuminated at normal incidence. The field is then

110 laterally uniform, only the zero-lateral-wavenumber (specular) component of
 111 the lattice coupling acts, and Eq. (2) becomes exactly a chain of n planes
 112 coupled by the specular lattice sums

$$\mathbf{S}(m) = \sum_{\mathbf{R}} \mathbf{G}(\mathbf{R} + md\hat{\mathbf{z}}), \quad (4)$$

113 where the sum runs over the lattice vectors $\mathbf{R} = d(0, p, q)$, p, q integers, that
 114 join the centres of the voxels in one plane (a square lattice of pitch d), and
 115 m is the separation of the two planes in units of d (Fig. 1a,b). The question
 116 is whether, and how fast, this chain converges to the continuous layer as
 117 $n \rightarrow \infty$ at fixed D .

118 *Implementations.* Every result is derived symbolically in Mathematica and
 119 compared with an independent numerical implementation or an exact solu-
 120 tion. “The package” below is the Python implementation of the voxel scheme
 121 (`cubic_scattering`): its cube T -matrix, its lattice kernels and its stratified
 122 propagator, which serve both as the object under test and, where validated,
 123 as a second implementation.

124 3. The continuous layer and its thin-layer expansion

125 At normal incidence a plane force drives a one-dimensional problem per
 126 wave type, with modulus M (M_P for P, μ for S), wavenumber k and plane
 127 Green’s function $g(z) = \frac{i}{2Mk} e^{ik|z|}$. The layer’s reflection and transmission
 128 follow from continuity of displacement and traction at its faces; they agree
 129 with the stratified propagator used as the reference in this work to 2.5×10^{-8} ,
 130 P and S (`ContinuumLimit_Reference.wl`).

131 For a layer of thickness D with modulus M_1 and density ρ_1 in a back-
 132 ground M_0, ρ_0 , write $Z_j = \sqrt{M_j \rho_j}$ for the impedances, $\zeta = Z_1/Z_0$ for their
 133 ratio and $k_1 = \omega \sqrt{\rho_1/M_1}$ for the layer's wavenumber. The reflection coeffi-
 134 cient expands as

$$R = \frac{ik_1 D}{2}(\zeta - \zeta^{-1}) + \frac{(k_1 D)^2}{4}(\zeta^{-2} - \zeta^2) - \frac{i(k_1 D)^3}{24}(\zeta - \zeta^{-1})(3\zeta^{-2} + 2 + 3\zeta^2) \\ + \frac{(k_1 D)^4}{48}(\zeta^2 - \zeta^{-2})(3\zeta^2 - 2 + 3\zeta^{-2}) + O(D^5), \quad (5)$$

135 exact in the contrast (`ContinuumLimit_ThinLayer.wl`, which checks each
 136 coefficient symbolically and the truncated series against the exact layer).
 137 The first term is

$$\frac{i\omega D}{2} \sqrt{\frac{M_0}{\rho_0}} \left[\frac{\rho_1 - \rho_0}{M_0} - \rho_0 \left(\frac{1}{M_1} - \frac{1}{M_0} \right) \right] : \quad (6)$$

138 to first order a thin layer is an interface across which the traction jumps by
 139 the excess inertia and the displacement by the excess compliance. At oblique
 140 incidence the same statement holds with the first-order system $\partial_z \mathbf{b} = \mathbf{A} \mathbf{b}$
 141 for the P-SV state $\mathbf{b} = (u_x, u_z, t_{xz}, t_{zz})$: the layer acts to $O(D)$ as the jump
 142 $\mathbf{I} + (\mathbf{A}_1 - \mathbf{A}_0)D$, which reproduces the layer's $O(D)$ reflection coefficients
 143 R_{PP}, R_{PS} to 10^{-12} ; the series truncated at $O(D^k)$ converges at exactly order
 144 k ; and a soft thin layer with shear modulus $\mu_1 = D/\eta$ at fixed compliance η
 145 tends to the linear-slip interface as $D \rightarrow 0$. These expansions are the target
 146 a single plane of voxels must reproduce order by order. A plane of voxels
 147 that carry Legendre moments of their internal field to degree p reproduces
 148 Eq. (5) term by term through $O(D^{2p+2})$ and first differs at D^{2p+3} (§8): the
 149 mean-only voxel reproduces the equivalent interface and the D^2 term, the
 150 first-moment voxel all four terms shown.

151 4. The specular sums of the point Green's tensor

152 *Why transform the sum.* The specular sum (4) adds, at a receiver, the field
 153 of every voxel of a plane m planes away. In that form it is a poor object to
 154 study. The Green's tensor decays slowly with distance, so the sum converges
 155 slowly, and it hides the one question that matters here: which part of it is
 156 the continuum, and which part is the discretisation. The Poisson summation
 157 formula [10] answers both at once, and is the standard tool for lattice sums
 158 of wave fields [11]. For a function f on the plane and the square lattice of
 159 pitch d it states

$$\sum_{\mathbf{R}} f(\mathbf{R}) = \frac{1}{d^2} \sum_{\mathbf{g}} \hat{f}(\mathbf{g}), \quad \hat{f}(\mathbf{g}) = \int f(\mathbf{x}) e^{-i\mathbf{g} \cdot \mathbf{x}} d^2x, \quad \mathbf{g} = \frac{2\pi}{d}(p, q), \quad (7)$$

160 with p, q integers: a sum of samples of f on the lattice equals a sum of samples
 161 of its Fourier transform on the reciprocal lattice, whose pitch is $2\pi/d$ (Fig. 2).
 162 Applied to the specular sum,

$$\mathbf{S}(m) = \frac{1}{d^2} \sum_{\mathbf{g}} \hat{\mathbf{G}}(\mathbf{g}; md), \quad (8)$$

163 with $\hat{\mathbf{G}}(\mathbf{g}; z)$ the two-dimensional (lateral) transform of the Green's tensor
 164 between planes a distance z apart. Each term has a physical meaning. Trans-
 165 formed laterally, the field of a plane of sources is a superposition of plane
 166 waves, one for each lateral wavenumber, and the lattice selects the wavenum-
 167 bers $\mathbf{g}$. The order $\mathbf{g} = 0$ is the laterally uniform plane wave that a continuous
 168 layer radiates: it is the continuum's term. Every other order has $|\mathbf{g}| \geq 2\pi/d$,
 169 far beyond the wavenumber of the field when the voxels are small compared
 170 with the wavelength, so it is evanescent and decays away from the plane as

171 $e^{-|\mathbf{g}|md} = e^{-2\pi m\sqrt{p^2+q^2}}$, independently of d . The transform therefore splits
 172 the lattice sum exactly into the continuum and a discretisation correction,
 173 and the correction converges after a few orders.

174 *The local term that survives.* In the strain-from-moment block the evanescent
 175 orders carry two powers of $|\mathbf{g}| \propto 1/d$, so their contribution per voxel volume
 176 does *not* vanish as $d \rightarrow 0$: at $m = 1$ it is eleven times the continuum
 177 term (`ContinuumLimit_SpecularSums.wl`, matched to the package’s Ewald
 178 lattice sum to 5×10^{-13}). This surviving term is static. Summed over all
 179 planes in slab order, the static lattice sum of the point kernel is exactly
 180 the depolarisation of a uniform plate ($-1/M_P$ on $\varepsilon_{zz} \leftarrow M_{zz}$ and $-1/(2\mu)$
 181 on the normal shears, per unit volume) plus a cubic-symmetric remainder
 182 (`ContinuumLimit_StaticLocal.wl`, to 2×10^{-10}). The point kernel therefore
 183 reproduces the continuum only if that remainder is cancelled by the voxel’s
 184 own self term.

185 5. The tiling identity

186 Replace the point coupling by the *source-cell average* $\langle \mathbf{G} \rangle(\mathbf{r}) = d^{-3} \int_{\text{cube}} \mathbf{G}(\mathbf{r}-$
 187 $\mathbf{y}) d\mathbf{y}$, the field at a voxel centre of a uniform polarisation of the source voxel.
 188 Each Poisson order in Eq. (8) is then multiplied by the cube’s form factor

$$F(\mathbf{g}) = \text{sinc}(g_x h) \text{sinc}(g_y h), \quad (9)$$

189 which vanishes at every $\mathbf{g} \neq 0$ of the reciprocal lattice since $\text{sinc}(p\pi) = 0$ for
 190 integer $p \neq 0$. This is the Fourier form of the fact that cubes tile the plane,
 191 $\sum_{\mathbf{R}} \chi(\mathbf{x} - \mathbf{R}) = 1$ for the cube’s indicator χ : a partition of unity (Fig. 3).
 192 Two exact statements follow.

193 *Between planes.* For $m \neq 0$ every evanescent order is removed and

$$\mathbf{S}_{\text{avg}}(m) = \frac{1}{d^2} \sum_{W=P,S} \hat{\mathbf{G}}_W(0; md) \text{sinc}(k_W h), \quad (10)$$

194 the continuum's plane-wave term times the vertical form factor. The pack-
195 age's cell-averaged kernel agrees in all 81 entries to 2.8×10^{-16} (`ContinuumLimit_`
196 `AveragedSums.wl`).

197 *Within a plane.* The sum over *all* cells of the plane, the voxel's own included,
198 is the field at the mid-plane of a uniform plate of thickness d (Fig. 1c):

$$d^3 \mathbf{S}_{\text{avg}}(0) + \mathbf{S}_{\text{self}} = d^3 \mathbf{P}_0, \quad \mathbf{S}_{\text{self}} = \int_{\text{cube}} \mathbf{G}(-\mathbf{y}) d\mathbf{y}, \quad (11)$$

199 where $\mathbf{P}_0 = d^{-3} \int_{-h}^h \mathbf{G}_{\text{plate}}(-z') dz'$ is elementary: in the one-dimensional
200 transform each entry is $c_0 + A/(Mk_z^2 - \rho\omega^2)$, so it averages to $c_0/d + A(e^{ikh} -$
201 $1)/(dMk^2)$. Its only nonzero entries are $u \leftarrow f$ on the diagonal and the
202 three normal strain-from-moment entries, whose local weights $-1/M_P$ and
203 $-1/(2\mu)$ are the plate depolarisation of §4. Statically the identity holds for
204 all 21 independent components of the strain-from-moment tensor to 4×10^{-9} ,
205 with every cell integral evaluated analytically (below).

206 *Consequence: an exact kernel with no lattice sum.* At normal incidence the
207 same-plane block of the cell-averaged Foldy–Lax kernel is

$$\mathbf{S}_{\text{avg}}(0) = \mathbf{P}_0 - \mathbf{S}_{\text{self}}/d^3, \quad (12)$$

208 with no Ewald splitting, near shell or asymptotic tail. The route it replaces
209 (Ewald summation with a near shell averaged by quadrature and an $O(d^2)$
210 Taylor tail beyond it) carries a static strain-from-moment error of 9.4×10^{-4}

211 at its default settings, falling to 2.6×10^{-5} only when both the shell radius
 212 and the quadrature order are raised; Eq. (12) is now the package's kernel at
 213 normal incidence.

214 *Cell integrals without quadrature.* Write the static (Kelvin) tensor as $G_{bc} =$
 215 $(c_A + c_B) \delta_{bc}/r - c_B \partial_b \partial_c r$ with $c_B = (1/\mu - 1/M_P)/8\pi$ and $c_A = 1/(4\pi\mu) - c_B$.
 216 Every cell integral of its derivatives is a cell integral of derivatives of $1/r$ and
 217 r . One application of the divergence theorem moves it onto faces $x_p = c_p \pm h$,
 218 which never contain the origin, and there the two-dimensional antiderivative
 219 in the face coordinates is continuous, so the face integral is a sum over the
 220 face's corners. The distributional term at $r = 0$ is carried by the divergence
 221 theorem itself. (A triple antiderivative evaluated at the cube's corners is
 222 *not* usable for repeated derivatives: its arctangent terms jump across the
 223 coordinate planes through the origin, and differentiating twice there discards
 224 exactly that term.) The self term so obtained reproduces the classical closed
 225 form of the cube's depolarisation to machine precision.

226 6. The single site: the cube T -matrix from moment integrals

227 Every error of a voxel scheme is made at the single site (§11), so the single
 228 site is derived here rather than imported. The classical route imports it. For
 229 an ellipsoid, Eshelby's uniformity theorem makes the internal strain under a
 230 uniform load uniform, with a depolarisation tensor in closed form [8, 12, 13];
 231 its dynamic generalisation is known for spheres and ellipsoids [14, 15]. For
 232 a cube or a general polyhedron the static field of a uniform eigenstrain is
 233 also in closed form [16, 17, 18, 19], but the strain it produces is not uniform,
 234 so there is no exact uniform-strain inclusion to import. In scattering theory

235 the same object appears as the self-consistent or effective-field closure of
 236 the volume integral equation [20, 21, 22]: a first-order (Born) scatterer [23]
 237 renormalised by the geometric series of its own near field, the step that turns
 238 a contrast into an effective contrast. Nestor [9] introduced the single site
 239 in this form, summing the series of the cell's self integral, with the singular
 240 near field evaluated over an infinitesimal cell. The moments below evaluate
 241 the same self integral over the finite cube, as a distribution. The elastic T -
 242 matrix of Waterman [24] represents a scatterer in spherical multipoles, the
 243 natural basis for a sphere; a voxel in a lattice needs one compatible with the
 244 lattice, and the moments over the cube provide it. The construction below
 245 makes the approximation of a voxel explicit rather than implicit: it derives
 246 the cube's T -matrix from the moments of the elastodynamic Green's tensor
 247 over the cube, and identifies the uniform-strain closure as one truncation of
 248 an exact hierarchy.

249 *6.1. The closed hierarchy*

250 Let the voxel be $V = [-h, h]^3$ with contrast $\delta c_{ijkl} = \Delta\lambda\delta_{ij}\delta_{kl} + \Delta\mu(\delta_{ik}\delta_{jl} +$
 251 $\delta_{il}\delta_{jk})$ and $\Delta\rho$. Expand the internal displacement about the centre to second
 252 degree and project the Lippmann–Schwinger equation onto 1, x_p and x_px_q .
 253 The unknowns are $u_i(0)$, $\partial_p u_i(0)$ and $\partial_p \partial_q u_i(0)$, and the result is the closed

254 system

$$(\delta_{ij} - \omega^2 \Delta \rho \mathbf{G}_{ij}) u_j + (\mathbf{N}_{in,k}^r \delta c_{nksj} - \frac{1}{2} \omega^2 \Delta \rho \mathbf{K}_{ij}^{rs}) \partial_r \partial_s u_j = u_i^0, \quad (13)$$

$$(\delta_{pr} \delta_{ij} + \omega^2 \Delta \rho \mathbf{N}_{ij,p}^r - \mathbf{M}_{in,pk} \delta c_{nkrj}) \partial_r u_j = \partial_p u_i^0, \quad (14)$$

$$-\omega^2 \Delta \rho \mathbf{M}_{ij,pq} u_j + (\delta_{qr} \delta_{ps} \delta_{ij} + \mathbf{Q}_{in,pqk}^r \delta c_{nksj} - \frac{1}{2} \omega^2 \Delta \rho \mathbf{P}_{ij,pq}^{rs}) \partial_r \partial_s u_j = \partial_p \partial_q u_i^0, \quad (15)$$

255 all fields evaluated at the centre. Its coefficients are six moments of the
256 background Green's tensor $\mathbf{G}$ over the voxel,

$$\begin{aligned} \mathbf{G}_{ij} &= \int_V G_{ij} dV, & \mathbf{K}_{ij}^{rs} &= \int_V G_{ij} x_r x_s dV, & \mathbf{N}_{in,k}^r &= \int_V \partial'_k G_{in} x_r dV, \\ \mathbf{M}_{in,pk} &= \int_V \partial'_p \partial'_k G_{in} dV, & \mathbf{P}_{ij,pq}^{rs} &= \int_V \partial'_q \partial'_p G_{ij} x_r x_s dV, & \mathbf{Q}_{in,pqk}^r &= \int_V \partial'_q \partial'_p \partial'_k G_{in} x_r dV, \end{aligned} \quad (16)$$

257 with derivatives taken with respect to the source point, $\partial' = -\partial$. That sign
258 reaches $\mathbf{N}$ and $\mathbf{Q}$ only, and no symmetry, parity or scaling check can detect
259 it, so it is fixed by convention. The moments are graded by (D, W) , the
260 number of derivatives and the degree of the weight: $\mathbf{G}(0, 0)$, $\mathbf{K}(0, 2)$, $\mathbf{N}(1, 1)$,
261 $\mathbf{M}(2, 0)$, $\mathbf{P}(2, 2)$, $\mathbf{Q}(3, 1)$.

262 6.2. The moments as distributions

263 Write the elastodynamic tensor as

$$G_{in}(\mathbf{r}) = \frac{1}{4\pi\mu} \left[\delta_{in} g_S(r) + \frac{1}{k_S^2} \partial_i \partial_n (g_S(r) - g_P(r)) \right], \quad g_c(r) = \frac{e^{ik_c r}}{r} \quad (17)$$

264 [25, 26]. Expanding g_c in powers of r reduces every moment in Eq. (16) to a
265 finite sum of scalar moments

$$E[m; d_1 \dots d_D; w_1 \dots w_W] = \langle \partial_{d_1} \dots \partial_{d_D} r^m, x_{w_1} \dots x_{w_W} \mathbf{1}_V \rangle. \quad (18)$$

266 For $m = -1$ and $D \geq 2$ the classical integrand is not absolutely integrable.
 267 $\partial^2(1/r)$ decays as r^{-3} , its angular average vanishes, and the value of the inte-
 268 gral depends on the shape of the region excluded around the origin. This is
 269 the source of the δ -function term of the inclusion problem, which a numerical
 270 quadrature misses. Without it the depolarisation is wrong; for a sphere it
 271 vanishes altogether (§6.3). It is not added afterwards here. The derivatives
 272 are distributional from the start, and each is moved onto the voxel's faces by
 273 pairing with the test function $w\mathbf{1}_V$, using $\partial_q\mathbf{1}_V = -n_q\delta_{\partial V}$:

$$\langle \partial_q T, w\mathbf{1}_V \rangle = \oint_{\partial V} n_q w T \, dA - \langle T, (\partial_q w) \mathbf{1}_V \rangle. \quad (19)$$

274 The surface term is evaluated on the faces, a distance h from the origin, where
 275 every derivative of r^m is smooth. The recursion lowers D and W together
 276 and ends on convergent integrals. The face integrals are done in closed form
 277 by the corner sums of §5. The sharp check is the sum rule

$$\sum_p E[-1; p, p;] = \nabla^2(1/r) \text{ paired with } \mathbf{1}_V = -4\pi, \quad (20)$$

278 which the face route satisfies exactly and a spherical excision returns as
 279 0. The closed forms contain logarithms of the units $(2 \pm \sqrt{3})^n$ of the Pell
 280 equation $x^2 - 3y^2 = 1$. Left unreduced, these cancel catastrophically: for
 281 example, $\log(72010600134783751 - 41575339372323900\sqrt{3})$ is a cancellation
 282 of 35 digits. They are therefore reduced to $n \log(2 - \sqrt{3})$, which is exact.
 283 Because every power of r is paired this way, the dynamic moments keep the
 284 even-power terms of the expansion $(k^2 r, r^3, \dots)$ as well as the odd ones.
 285 A route that integrates only the polynomial (odd-power) terms of $\mathbf{G}$ misses
 286 those terms, at a real $O((kh)^2)$ cost (§6.4).

287 6.3. Parity, and the nine-component block

288 The cube is centrosymmetric. A moment that couples a field quantity of
 289 degree a to one of degree b therefore vanishes unless $a + b$ is even. u , ∂u and
 290 $\partial\partial u$ have degrees 0, 1 and 2, so Eq. (14) contains the first gradient alone: ∂u is
 291 determined by its own block and couples to neither u nor $\partial\partial u$. The leading
 292 scattering by a stiffness contrast is the stress dipole $\int_V \delta c : \nabla u \, dV$, which
 293 needs only ∂u . The leading far field of the stiffness contrast is therefore fixed
 294 by Eq. (14), and *no treatment of the second-gradient block (15) can change*
 295 *it*. What Eq. (15) contributes reaches the far field only through the $O((kh)^2)$
 296 coupling in Eq. (13). That is the order, and the origin (the variation of the
 297 field across the voxel), of the collocation error of §7, which the first-moment
 298 voxel of §8 removes.

299 With $\Delta\rho = 0$, Eq. (14) is the 9×9 operator $\delta_{pr}\delta_{ij} - \mathbf{M}_{in,pk}\delta c_{nkrj}$, built
 300 from $\mathbf{M}$ alone. Under O_h the nine gradient components decompose as $A_{1g} \oplus$
 301 $E_g \oplus T_{1g} \oplus T_{2g}$, and the operator is diagonal in that basis. Its eigenvalues are
 302 the inverse strain concentrations:

$$A_{1g} \text{ (dilatation)} : \quad 1 + \frac{\Delta\lambda + \frac{2}{3}\Delta\mu}{\lambda + 2\mu}, \quad (21)$$

$$T_{2g} \text{ (off-diagonal shear)} : \quad 1 + 2\Delta\mu S_{\text{off}}, \quad S_{\text{off}} = \frac{\pi(\lambda + 2\mu) - \sqrt{3}(\lambda + \mu)}{3\pi\mu(\lambda + 2\mu)}, \quad (22)$$

$$E_g \text{ (diagonal shear)} : \quad 1 + 2\Delta\mu S_{\text{diag}}, \quad S_{\text{diag}} = \frac{3\sqrt{3}(\lambda + \mu) + 2\pi\mu}{6\pi\mu(\lambda + 2\mu)}. \quad (23)$$

303 The density channel is the displacement block of Eq. (13) with the second
 304 gradients dropped, an amplification $(1 - \omega^2\Delta\rho\mathbf{G})^{-1}$ with $\mathbf{G}$ the cube integral
 305 of $\mathbf{G}$.

306 *The cube against the sphere.* For a sphere, Eshelby's tensor gives a dilata-
 307 tional channel $1 + \Delta K / (\lambda + 2\mu)$, with $\Delta K = \Delta\lambda + \frac{2}{3}\Delta\mu$, and a single deviatoric
 308 channel $1 + 2\Delta\mu S_{\text{sph}}$, with $2\mu S_{\text{sph}} = 2(4 - 5\nu)/[15(1 - \nu)]$. The cube's di-
 309 latational channel, Eq. (21), is the sphere's exactly: its eigenvalue involves
 310 $\mathbf{M}$ only through the contraction $\int_V \partial_i \partial_n G_{in} dV$, and the static Navier equa-
 311 tion makes $\partial_i \partial_n G_{in} = -\delta(\mathbf{r})/(\lambda + 2\mu)$, whose integral does not depend on
 312 the shape. The cube's shear channels differ from each other. Their isotropic
 313 average, however, is the sphere's, also exactly:

$$\frac{1}{5}(3S_{\text{off}} + 2S_{\text{diag}}) = S_{\text{sph}}, \quad S_{\text{diag}} - S_{\text{off}} = \frac{(\lambda + \mu)(5\sqrt{3} - 2\pi)}{6\pi\mu(\lambda + 2\mu)} > 0. \quad (24)$$

314 The $\sqrt{3}$ terms cancel in the average. What a cube adds to a sphere at this
 315 order is therefore a pure cubic anisotropy. It splits the deviatoric channel
 316 about the sphere's value, weighted 3 : 2 by the dimensions of T_{2g} and E_g ,
 317 and the diagonal shear is the more depolarised. At the background of §7
 318 ($\nu = 7/32$), $2\mu S_{\text{off}} = 0.4314$, $2\mu S_{\text{diag}} = 0.5929$ and the sphere's value is
 319 0.4960.

320 *Where Eshelby enters.* Equations (21)–(23) follow from the moment integrals
 321 and the group theory alone. No ellipsoidal result is used, and no δ term is
 322 added by hand. What corresponds to Eshelby is the *truncation*. Keeping
 323 ∂u and discarding $\partial\partial u$ is exactly the uniform-internal-strain ansatz, and by
 324 the uniformity theorem that ansatz is exact if and only if the inclusion is
 325 an ellipsoid. For a sphere the construction is therefore exact in the static
 326 limit, and for a cube it is an approximation. The hierarchy (13)–(15) states
 327 that approximation as the omitted blocks, rather than as an unquantified
 328 shape correction. A polynomial expansion about the centre cannot represent

329 the edge and corner concentrations of an isolated cube's internal field at any
330 finite degree. A voxel in a uniform layer has no such concentrations: its
331 neighbours have the same properties, so its internal field is the continuum's,
332 which is smooth. That is why a low-degree internal field suffices in a lattice
333 (§§7–8) although it is only an approximation for an isolated cube.

334 *Against the exact sphere.* These statements are tested against the exact elas-
335 tic Mie solution for a sphere, with no Eshelby formula in between. The
336 concentration of Mie order n is $E_n = a_n/a_n^{\text{Born}}$ (orders 0, 1, 2 are the dilata-
337 tional, density and shear channels), extrapolated to $ka \rightarrow 0$. The formulation
338 is applied to a ball of radius a , with its moment $\mathbf{M}$ evaluated by the surface
339 route of Eq. (19) on $|\mathbf{r}| = a$. Its dilatational and shear channels equal Mie's
340 to 2×10^{-10} , and the density channel is 1. The cube's closed forms meet
341 the same numbers. Its dilatational channel is Mie's E_0 to 2×10^{-10} , and its
342 isotropic shear average, Eq. (24), is the depolarisation read from Mie's E_2
343 to 10^{-8} . The same moment taken as a principal value, over the ball with
344 a small sphere excluded around the origin, has zero depolarisation in every
345 channel. The whole static response of the sphere is the δ term.

346 At finite frequency the closure uses the dynamic ball moments, which are
347 in closed form. It departs from Mie at second order in ka and first order in
348 the contrast, and the departure is closed-form too: the ball's moments are
349 elementary, and Mie's low-frequency series with the contrast kept exact gives

350 E_n to relative order $(ka)^3$. To first order in the contrast,

$$\frac{E_n^{\text{closure}}}{E_n^{\text{Mie}}} - 1 = \begin{cases} -\frac{(k_P a)^2}{10} \frac{\Delta\lambda}{\lambda + 2\mu}, & n = 0, \\ \frac{2(k_S a)^2 + (k_P a)^2}{30} \frac{\Delta\rho}{\rho}, & n = 1, \\ -\frac{3(k_S a)^2 + 2(k_P a)^2 \beta^2 / \alpha^2}{75} \frac{\Delta\mu}{\mu}, & n = 2, \end{cases} \quad (25)$$

351 with no $O(ka)$ term and no $O((ka)^3)$ term. The imaginary $(ka)^3$ terms,
 352 the radiation damping, agree exactly in all three channels. So the closure's
 353 only error at this order is the real $(ka)^2$ term, the uniform-field ansatz's
 354 own. That is the order the hierarchy assigns to the omitted second gradi-
 355 ents (§6.3), and it has the same origin, the variation of the field across the
 356 scatterer, as the collocation error of §7. Eq. (25) is derived symbolically
 357 (`SphereClosureDynamic.wl`). It agrees with the coefficient extracted from
 358 the numerical Mie solve and an independent implementation of the closure
 359 to 5×10^{-4} , which is the size of the next order in the contrast.

360 6.4. The collocation closure

361 The collocation closure takes the internal field of a voxel as uniform, the
 362 truncation of §6.3 for the density and stiffness channels together. The voxel's
 363 own field at its centre is then $\mathbf{S}_{\text{self}} \Delta e$ for internal state e , where $\mathbf{S}_{\text{self}}$ is the
 364 nine-component form of the moments $\mathbf{G}$ and $\mathbf{M}$. By parity it is block-diagonal.
 365 So $e = \psi + \mathbf{S}_{\text{self}} \Delta e$ and

$$\mathbf{T} = d^3 \Delta (\mathbf{I} - \mathbf{S}_{\text{self}} \Delta)^{-1}. \quad (26)$$

366 With the kernel (12) the self term cancels. Substituting $\psi = (\mathbf{I} - \mathbf{S}_{\text{self}}\Delta)e$
 367 into Eq. (2),

$$e_i = \psi_i^0 + d^3 \sum_j \mathbf{K}_{\text{all}}(i, j) \Delta e_j, \quad \mathbf{K}_{\text{all}}(i, i) = \mathbf{P}_0, \quad \mathbf{K}_{\text{all}}(i, j) = \mathbf{S}_{\text{avg}}(i - j), \quad (27)$$

368 an identity checked on the matrices to 4×10^{-16} . Every block of $\mathbf{K}_{\text{all}}$ is an
 369 *exact* cell integral of the one-dimensional continuum kernel (Eqs. (10), (11)),
 370 so Eq. (27) is precisely the collocation of the continuum Lippmann–Schwinger
 371 equation with a strain piecewise constant over voxels. It does not contain
 372 $\mathbf{S}_{\text{self}}$ at all.

373 The package’s cube T -matrix is this closure statically: both blocks agree
 374 with Eq. (26) to 2×10^{-5} at $k_S h = 0.01$, the difference falling as $(kh)^2$
 375 (`ContinuumLimit_CubeT.w1`). It departs dynamically by about $0.25(kh)^2$
 376 through its far-field form-factor corrections. Its self integrals are exact in the
 377 static and radiation parts but omit the real even-power terms of the Green’s
 378 tensor’s expansion ($k^2 r, r^3, \dots$), an error of 0.4–0.6 $(kh)^2$ against the analytic
 379 self term, whose series agrees with the closed-form elastodynamic tensor to
 380 10^{-14} . By Eq. (27) that error cannot enter the closure scheme.

381 7. Convergence, and the closed-form error

382 *Second order.* For a layer $D = 2\text{ m}$ with the contrast $\Delta\lambda = 2\text{ GPa}$, $\Delta\mu =$
 383 1 GPa , $\Delta\rho = 100\text{ kg m}^{-3}$ in a background $\alpha = 5000$, $\beta = 3000\text{ m s}^{-1}$, $\rho =$
 384 2500 kg m^{-3} , illuminated by a P plane force, the reflected displacement of
 385 the chain converges to the continuous layer’s at order 2.000 for $n = 4 \rightarrow$
 386 16 at $\omega = 60, 300$ and 600 rad s^{-1} , with the collocation (27) and with the

387 package *T*-matrix alike (`ContinuumLimit_Chain.wl`). The error depends on
 388 $k_S h$ alone.

389 *The leading error in closed form.* In Eq. (27) the kernel is integrated exactly;
 390 the only approximation is the internal field's variation across a voxel. At first
 391 order in the contrast the discrete scattered field is therefore a midpoint rule
 392 applied to the continuum's Born integral, with the kernel's cell integral exact.
 393 For reflection the integrand carries the two-way phase e^{2ikz} ; for transmission
 394 it is flat. With k the background wavenumber of the wave concerned (k_P for
 395 an incident P wave, k_S for an incident S wave), R the reflection coefficient
 396 and T^{sc} the scattered part of the transmitted field, for *every* n

$$\frac{R_{\text{disc}}}{R} = \frac{\text{sinc}(kh)}{\text{sinc}(2kh)} = 1 + \frac{(kd)^2}{8} + O(d^4), \quad \frac{T_{\text{disc}}^{\text{sc}}}{T^{\text{sc}}} = \text{sinc}(kh) = 1 - \frac{(kd)^2}{24} + O(d^4), \quad (28)$$

397 confirmed for $n = 1$ to 16 to 10^{-4} of the error itself (`ContinuumLimit_`
 398 `ErrorConstant.wl`); for an incident S wave, with $k = k_S$, the same forms
 399 and those of Eq. (31) hold to 4×10^{-4} (`ContinuumLimit_IncidentS.wl`, §8).
 400 At the full contrast the reflection constant is unchanged (0.124 against 1/8);
 401 the transmission constant grows to -0.092 .

402 *Why a uniform-field single site stops at second order.* The two constants have
 403 opposite signs, so no scalar correction of $\mathbf{T}$, no single $O((kh)^2)$ factor, can
 404 cancel both. The package *T*-matrix illustrates it: its form-factor corrections
 405 reduce the reflection error by a factor 4.45 and *increase* the transmission
 406 error by 1.7, at every n . The missing term in Eq. (28) is the first moment
 407 of the internal field over each voxel, which radiates through the derivative of
 408 the kernel. A fourth-order scheme must carry it: a single site with a gradient

channel, and a kernel whose cell integrals include the kernel's derivative.

8. Fourth order and beyond: Legendre moments of the voxel field

Let each voxel carry the mean *and* the first moment of its internal field. On voxel j , with local coordinate $s \in [-h, h]$ and Legendre functions $\phi_b(s) = P_b(s/h)$ ($\phi_0 = 1$, $\phi_1 = s/h$, $\phi_2 = (3s^2/h^2 - 1)/2$, ...), the field is $w = \sum_b c_{b,j} \phi_b$, $w = (u, \varepsilon)$ the displacement and strain of the P channel, and the polarisation is $q = (\omega^2 \Delta \rho u, \Delta M \varepsilon)$ with $\Delta M = \Delta \lambda + 2\Delta \mu$. Testing the continuum equation

$$w(z) = w^0(z) + \int \mathbf{K}(z-z') q(z') dz', \quad \mathbf{K} = \begin{pmatrix} g & g' \\ g' & g'' \end{pmatrix}, \quad g'' = -k^2 g - \delta(z)/M_P, \quad (29)$$

with the same functions on every voxel gives

$$\left(\int_i \phi_a^2 dz \right) c_{a,i} = \int_i \phi_a w^0 dz + \sum_{j,b} \left[\int_i \int_j \phi_a(z) \mathbf{K}(z-z') \phi_b(z') dz dz' \right] \Delta_1 c_{b,j}, \quad (30)$$

$\Delta_1 = \text{diag}(\omega^2 \Delta \rho, \Delta M)$. Between voxels the double integral is separable, a product of Legendre moments of $e^{\pm i k z}$; within a voxel it is split at $z' = z$ and the local term contributes $-\int_i \phi_a \phi_b dz / M_P$. Every entry is closed-form. Keeping only ϕ_0 recovers a second-order scheme; keeping ϕ_0 and ϕ_1 converges at fourth order in both reflection and transmission (`ContinuumLimit_FourthOrder.wl`; Table 1). A single plane of first-moment voxels represents the 2 m layer 6500 times more accurately than a single plane of uniform ones. The collocation column is the scheme of §6 and reproduces the chain of the full 9×9 system to all printed digits.

Table 1: Relative error of the scattered displacement of the layer, reflection / transmission, at $\omega = 300 \text{ rad s}^{-1}$, against the number of voxel planes n : collocation, the mean-only voxel and the first-moment voxel of Eq. (30). The order is 4.00 at $\omega = 60$ and 600 rad s^{-1} too.

n	collocation	mean only	mean and first moment
1	$1.8 \times 10^{-3} / 1.3 \times 10^{-3}$	$1.2 \times 10^{-3} / 1.8 \times 10^{-3}$	$2.8 \times 10^{-7} / 2.8 \times 10^{-7}$
4	$1.1 \times 10^{-4} / 8.3 \times 10^{-5}$	$7.5 \times 10^{-5} / 1.1 \times 10^{-4}$	$1.1 \times 10^{-9} / 1.1 \times 10^{-9}$
16	$7.0 \times 10^{-6} / 5.2 \times 10^{-6}$	$4.7 \times 10^{-6} / 6.9 \times 10^{-6}$	$4.2 \times 10^{-12} / 4.2 \times 10^{-12}$
order	2 / 2	2 / 2	4 / 4

427 *Higher moments.* Carrying the Legendre moments to degree p changes nothing in the construction: the moments and the same-cell double integrals of
428 $P_b(s/h)$ against exponentials are again closed-form, and for $p \leq 1$ the general-
429 degree solver reproduces the chain above exactly. The second-moment voxel
430 ($p = 2$) converges at sixth order and the third-moment voxel ($p = 3$) at
431 eighth, in reflection and transmission (Table 2; `ContinuumLimit_SecondMoment`.
432 `w1`); the second-moment voxel's order is 6.00 at $\omega = 60, 300$ and 600 rad s^{-1} .
433 The errors fall to 10^{-22} , so they are computed in 40-digit arithmetic.

435 *The leading error in closed form.* Without scattering, the Galerkin solution
436 is the L^2 projection Π of the incident field onto each voxel's basis, and the
437 kernel is integrated over each voxel exactly; at first order in the contrast the
438 error is therefore the kernel's cell integral applied to $\Pi w^0 - w^0$. The field
439 varies as e^{iks} across a voxel and the kernel toward an outside observer as e^{iqs} ,
440 $q = +k$ for reflection and $q = -k$ for transmission, so for every scheme and

Table 2: Relative error of the scattered displacement of the layer, reflection / transmission, at $\omega = 300 \text{ rad s}^{-1}$, for voxels carrying Legendre moments to degree p . The order is the mean over the halvings $n = 2 \rightarrow 4 \rightarrow 8$.

n	first moment ($p = 1$)	second moment ($p = 2$)	third moment ($p = 3$)
1	$2.8 \times 10^{-7} / 2.8 \times 10^{-7}$	$2.8 \times 10^{-11} / 4.2 \times 10^{-11}$	$1.5 \times 10^{-15} / 1.5 \times 10^{-15}$
2	$1.7 \times 10^{-8} / 1.7 \times 10^{-8}$	$4.4 \times 10^{-13} / 6.5 \times 10^{-13}$	$6.0 \times 10^{-18} / 6.0 \times 10^{-18}$
4	$1.1 \times 10^{-9} / 1.1 \times 10^{-9}$	$6.9 \times 10^{-15} / 1.0 \times 10^{-14}$	$2.4 \times 10^{-20} / 2.4 \times 10^{-20}$
8	$6.7 \times 10^{-11} / 6.7 \times 10^{-11}$	$1.1 \times 10^{-16} / 1.6 \times 10^{-16}$	$9.2 \times 10^{-23} / 9.2 \times 10^{-23}$
order	4 / 4	6 / 6	8 / 8

every n

$$\frac{\text{discrete}}{\text{exact}} = \frac{\int_{-h}^h e^{iqs} \Pi[e^{iks}] ds}{\int_{-h}^h e^{i(q+k)s} ds}, \quad (31)$$

with Π the centre value for collocation, the cell mean for the mean-only voxel and the projection onto $\{P_0, \dots, P_p\}$ for the voxel of degree p (`ContinuumLimit_FourthOrderError.wl`, `ContinuumLimit_SecondMoment.wl`). The results are in Table 3. The chain of Eq. (30) reproduces the forms for collocation and degrees 0 and 1 at $n = 1$ to 8 to 10^{-4} of the error, the remainder being the second order in the contrast, and those for degrees 2 and 3 to 4×10^{-7} at a contrast reduced by 10^{-6} . The constant for degree p follows from the Legendre expansion $e^{iks} = \sum_j i^j (2j+1) j_j(kh) P_j(s/h)$. The projection drops the terms $j > p$, led by $j = p+1$, whose coefficient is $O((kh)^{p+1})$; its moment against e^{iqs} is $2h i^{p+1} j_{p+1}(qh) = O((qh)^{p+1})$. With $j_n(x) \approx x^n / (2n+1)!!$ and

Table 3: The leading error of the discrete layer in closed form, Eq. (31): discrete over exact scattered field at first order in the contrast, for every number of voxel planes n .

	reflection	transmission
collocation	$1 + (kd)^2/8 + O(d^4)$	$1 - (kd)^2/24 + O(d^4)$
mean only	$1 + (kd)^2/12 + O(d^4)$	$1 - (kd)^2/12 + O(d^4)$
first moment	$1 - (kd)^4/720 + O(d^6)$	$1 - (kd)^4/720 + O(d^6)$
second moment	$1 + (kd)^6/100800 + O(d^8)$	$1 - (kd)^6/100800 + O(d^8)$
third moment	$1 - (kd)^8/25401600 + O(d^{10})$	$1 - (kd)^8/25401600 + O(d^{10})$
degree p	$1 + (-1)^p c_p (kd)^{2p+2}$	$1 - c_p (kd)^{2p+2}$

452 $h = d/2,$

$$c_p = \frac{2p+3}{4^{p+1} [(2p+3)!!]^2} = \frac{[(p+1)!]^2}{(2p+2)! (2p+3)!}, \quad c_0 = \frac{1}{12}, \quad c_1 = \frac{1}{720}, \quad c_2 = \frac{1}{100800}, \quad (32)$$

453 and the leading error carries $(q/k)^{p+1}$: the same sign in reflection and trans-
454 mission for odd p , opposite signs for even p . The first-moment voxel's leading
455 error therefore does not depend on the direction in which the field is observed.
456 Eq. (32) is checked symbolically for $p = 0$ to 5 (every lower power of kd van-
457 ishes) in Mathematica and, independently, in sympy.

458 *One plane of voxels and the thin-layer series.* Refining at fixed thickness is
459 one limit; thinning a single plane is the other. For one plane of voxels ($n = 1$,
460 $d = D$) the series in D of the reflection coefficient, computed exactly at the
461 full contrast, coincides with that of the exact layer, Eq. (5), term by term
462 through D^{2p+2} and first differs at D^{2p+3} , for $p = 0$ to 3 and for a layer both
463 stiffer and softer than its surroundings (`ContinuumLimit_SecondMoment`).

464 w1). Each further Legendre moment adds two terms of the thin-layer series.

465 *In three dimensions.* The three-dimensional first-moment voxel carries the
 466 nine-component state with Legendre weights $\{1, x/h, y/h, z/h\}$, 36 unknowns
 467 in all, tested with the same functions and coupled by cell-to-cell double in-
 468 tegrals of the Green's tensor. For a laterally uniform layer at normal in-
 469 cidence it reduces *exactly* to Eq. (30) (ContinuumLimit_Reduction3D.w1).
 470 The lateral first moments are never excited: the lattice and the incident
 471 field are symmetric under reflection about every voxel's centre planes, and
 472 $x/h, y/h$ are odd there. The mean and the z -moment carry weights that
 473 are constant laterally, so their lateral form factor $\text{sinc}(g_x h) \text{sinc}(g_y h)$ van-
 474 ishes on the reciprocal lattice and the plane sum of the double integrals
 475 is the plate-to-plate integral in z alone. Two facts mark where genuinely
 476 three-dimensional structure begins. The lateral first moment's form fac-
 477 tor, $F_1(g) = i(gh \cos gh - \sin gh)/(gh)^2$, does not vanish on the recipro-
 478 cal lattice ($F_1 = i(-1)^p/p\pi$), so the gradient channels couple through the
 479 evanescent orders. And at oblique incidence the mean's own zeros shift,
 480 $\text{sinc}((2\pi p/d + k_x)h) = (-1)^p k_x h/p\pi + O(k_x^2 h^2)$, so the tiling identity holds
 481 only to $O(k_{\parallel} h)$.

482 *Oblique incidence.* The scheme is built spectrally. By Poisson summation
 483 the Bloch coupling between planes i and j at lateral wavenumber $\mathbf{k}_{\parallel}$ is

$$C_{ab}(i, j) = \frac{1}{d^2} \sum_{\mathbf{g}} L_a(\boldsymbol{\kappa}) L_b(-\boldsymbol{\kappa}) \iint \phi_a(z) \hat{\mathbf{G}}(\boldsymbol{\kappa}; z-z') \phi_b(z') dz dz', \quad \boldsymbol{\kappa} = \mathbf{k}_{\parallel} + \mathbf{g},$$

(33)

484 with L_a the lateral Legendre factors. The whole-space tensor decays as
 485 $1/k_z^2$ and the nine-component state adds at most two powers of k_z , so each

entry of $\hat{\mathbf{G}}(\boldsymbol{\kappa}; z)$ is a $\delta(z)$ of constant weight plus, per mode, $[\mathrm{i}\alpha/2\gamma + \mathrm{i}\beta \operatorname{sgn}(z)/2] \mathrm{e}^{\mathrm{i}\gamma|z|}$, $\gamma^2 = k_W^2 - \kappa^2$: every z integral is closed-form. Double averaging makes the lattice sum converge without Ewald splitting, to three digits at $|p|, |q| \leq 8$. Against the exact layer, for a P wave incident at 20° and 40° (`ContinuumLimit_Oblique.wl`; Table 4). The mean-only voxel is

Table 4: Relative error of the reflected P (R_{PP}) and converted S (R_{PS}) waves for a P wave incident at 20° , with the mean-only and first-moment voxels of Eq. (33). At 40° the orders are again 2 and 4.

n	mean only		mean and first moments	
	R_{PP}	R_{PS}	R_{PP}	R_{PS}
1	9.3×10^{-4}	1.7×10^{-3}	7.2×10^{-8}	4.1×10^{-7}
2	2.3×10^{-4}	4.2×10^{-4}	4.5×10^{-9}	2.5×10^{-8}
8	1.5×10^{-5}	2.7×10^{-5}	1.7×10^{-11}	9.9×10^{-11}
order	2	2	4	4

second order and the first-moment voxel fourth order in both the reflected P and the converted S wave: the breakdown of the tiling identity at $O(k_{\parallel}h)$ and the evanescent coupling of the gradient channels do not cost an order. At $\mathbf{k}_{\parallel} = 0$ the construction reproduces Eq. (30) to all printed digits.

Incident S. Nothing in the construction is specific to P. For a normally incident S wave the layer is the one-dimensional problem of Eq. (30) with $M_P \rightarrow \mu$, $k_P \rightarrow k_S$ and $\Delta M \rightarrow \Delta\mu$; the collocation and mean-only schemes converge at order 2.00 and the first-moment voxel at 4.00, in reflection and transmission, and the full 9×9 construction with an incident SH wave reproduces that one-dimensional result to all printed digits. At oblique inci-

501 dence, against the exact layer (`ContinuumLimit_IncidentS.wl`), the orders
are those of Table 5. The orders, and with $k = k_S$ the closed-form constants

Table 5: Observed order of convergence ($n = 2 \rightarrow 8$, $\omega = 300 \text{ rad s}^{-1}$) for incident SV and SH waves, with the mean-only and first-moment voxels. The SV angles lie below the critical angle $\arcsin(\beta/\alpha) = 36.9^\circ$.

	mean only	mean and first moments
SV at 20° : R_{SS} / R_{SP}	2.00 / 2.00	4.00 / 3.99
SV at 30° : R_{SS} / R_{SP}	1.98 / 2.00	3.97 / 4.00
SH at 20° and 40° : R_{SH}	2.00	4.00

502

503 of Eqs. (28) and (31), are those of the incident P wave.

504 The spectral construction is implemented twice. The second implementa-
505 tion (`scripts/crosscheck_first_moment_voxel.py`) shares only the defi-
506 nition of the scheme: it takes the depth dependence of the transformed kernel
507 from an independent residue construction, obtains the local weight as a nu-
508 merical large- k_z limit of the inverted Christoffel matrix, evaluates every dou-
509 ble integral by quadrature and separates P from S by sampling the upgoing
510 field at two depths. The raw reflected displacements of the two agree to $9 \times$
511 10^{-16} in all 36 cases compared: incident P, SV and SH, normal incidence and
512 20° , $n = 1, 2, 4$, with and without the first moments. The one-dimensional
513 scheme of any degree is also implemented twice. The second implementation
514 (`scripts/crosscheck_second_moment_voxel.py`) evaluates every integral
515 by Gauss quadrature, split at the receiver within a voxel, and the exact layer
516 from its own interface system, in double precision. Its errors agree with the
517 40-digit ones within its round-off floor (3×10^{-14} in reflection, 1.6×10^{-13} in

518 transmission); at $\omega = 1200 \text{ rad s}^{-1}$ it measures orders 4.02 / 3.99 and 6.02 /
 519 6.00; for one plane as the layer thins its differences from the exact layer fall
 520 as $D^{3.00}$, $D^{5.00}$ and $D^{7.00}$ for $p = 0, 1$ and 2 ; and it rederives c_p for $p \leq 5$.
 521 Figure 5(a,b) shows the convergence, with the orders fitted to it.

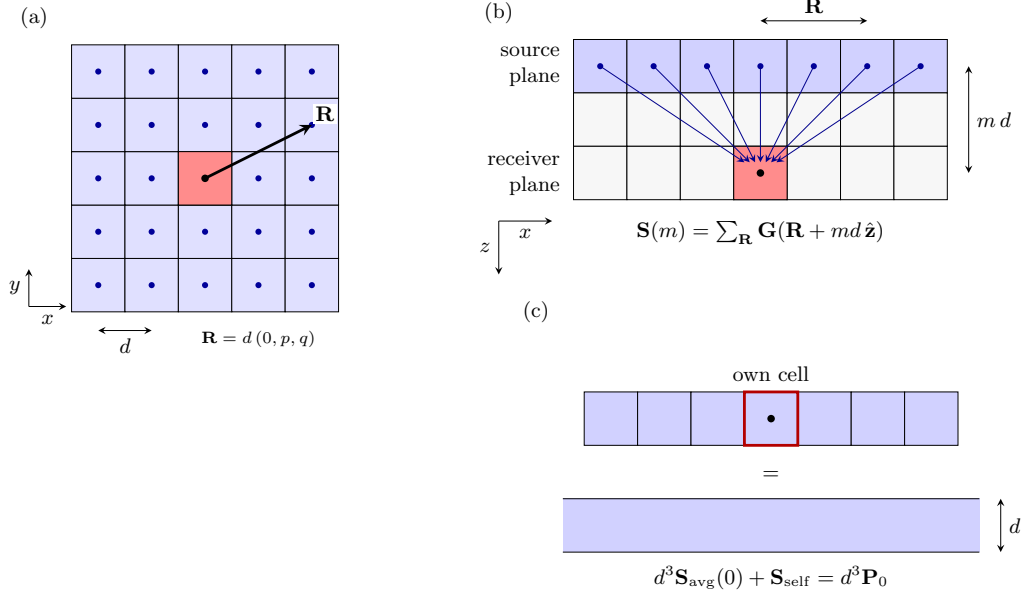


Figure 1: The lattice and its sums. (a) Plan view of one plane of voxels: their centres form a square lattice of pitch d , and a lattice vector $\mathbf{R} = d(0, p, q)$ joins the centre of one voxel (red) to another. (b) Cross-section (z down), between planes: the voxel of a receiver plane (red) is coupled to every voxel of a source plane m planes above it, one for each lattice vector $\mathbf{R}$; the sum over $\mathbf{R}$ is the specular lattice sum of Eq. (4). (c) Within a plane: the sum over every cell of the plane, the receiver's own cell (outlined) included, is the field of a uniform plate of thickness d , the tiling identity of Eq. (11).

Alt text: Three schematic panels. On the left, a plan view of a five-by-five grid of square voxels with a dot at each centre; the centre voxel is highlighted, an arrow labelled $\mathbf{R}$ runs from its centre to another voxel centre, and the spacing between neighbouring centres is marked d . At the top right, a cross-section through three rows of voxels: arrows from the centre of every voxel in the top row converge on a highlighted voxel two rows below, with the plane separation $m d$ marked. Below it, a single row of voxels with one outlined as the receiver's own cell is shown equal to a continuous shaded band of the same thickness d , a uniform plate.

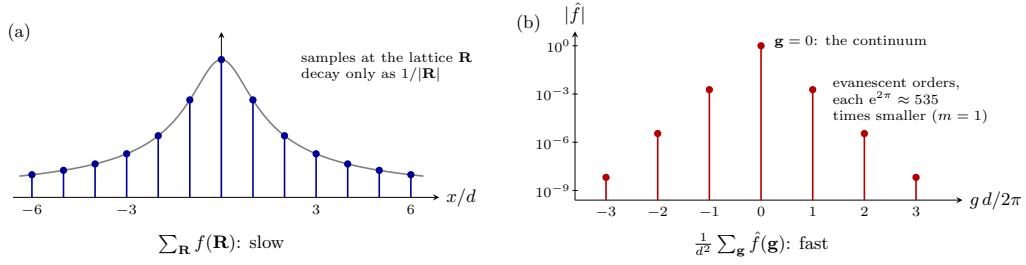


Figure 2: The Poisson summation formula (7) applied to a plane of voxels, along one lateral axis. (a) Real space: the field of each voxel of a source plane at a receiver one plane away, sampled at the lattice points; the samples decay only as $1/|\mathbf{R}|$, so their sum converges slowly. (b) The same sum after Poisson summation, on a logarithmic scale: samples of the lateral Fourier transform at the reciprocal-lattice orders $\mathbf{g} = 2\pi p/d$. The order $\mathbf{g} = 0$ is the continuum's plane wave; every other order is evanescent and, at $m = 1$, smaller than its neighbour by $e^{2\pi} \approx 535$. Magnitudes are schematic: the static decay $e^{-|\mathbf{g}|md}$ is shown.

Alt text: Two panels. On the left, a smooth bell-shaped curve that decays slowly on both sides is sampled by vertical stems at integer positions from minus six to six, the stems remaining clearly visible far from the centre. On the right, on a logarithmic vertical axis from one down to ten to the minus nine, a stem of height one at zero is labelled as the continuum, and stems at plus and minus one, two and three fall by about three powers of ten per step and are labelled as evanescent orders.

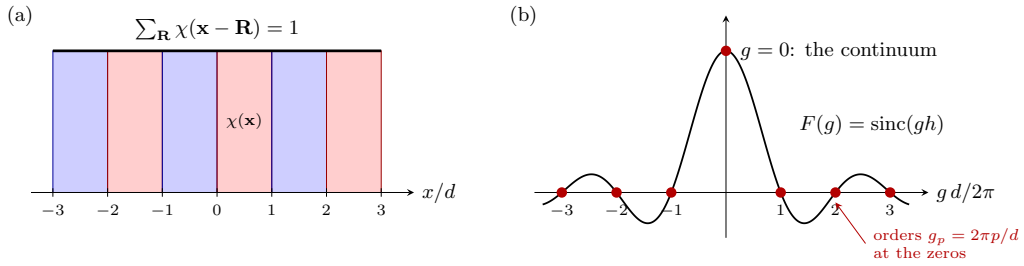


Figure 3: Why cubes make the coupling exact. (a) Cubes tile the plane: the indicator functions χ of the translated cells sum to exactly one everywhere, a partition of unity. (b) The same fact in Fourier space: the cube's lateral form factor $F(g) = \text{sinc}(gh)$ (shown along one axis) vanishes at every reciprocal-lattice order $g_p = 2\pi p/d$ with $p \neq 0$, so of all the orders in the lattice sum (8) only $g = 0$, the continuum's plane wave, survives the source-cell average.

Alt text: Two panels. On the left, adjacent box functions of unit height and unit width, alternately shaded, lie side by side along an axis marked in voxel widths, and a thick horizontal line at height one marks their sum. On the right, the sinc curve, equal to one at the origin and oscillating with decaying amplitude, is plotted against wavenumber in units of $2\pi/d$; dots mark the reciprocal-lattice orders, one at height one at the origin and all the others on the axis at the zeros of the curve.

522 9. A heterogeneous layer

523 A heterogeneous medium is not a layer of identical cells. The simplest
524 heterogeneous case with an exact answer is a stratified model built from layers
525 of constant properties, with real interfaces between them: the form of many
526 layered crustal velocity models. The last test takes such a model. A layer
527 $D = 4$ m thick is divided into eight model layers of 0.5 m, and in each one $\Delta\lambda$,
528 $\Delta\mu$ and $\Delta\rho$ are drawn uniformly within $\pm 15\%$ of the background (the values
529 are listed in `ContinuumLimit_Heterogeneous.w1`). The result is a random
530 stratification with physical contrasts. The voxels refine the model's layering,
531 with m planes of voxels in each model layer, so no voxel straddles a model
532 interface: the natural discretisation when the model's interfaces are known.
533 At $\omega = 1500$ rad s⁻¹ the layer is $k_P D = 1.2$ and $k_S D = 2.0$ thick, so the field
534 reverberates between the model interfaces. The exact answer is the product
535 of the model layers' propagators. Its same-type reflection coefficients agree
536 with those of the reflection–transmission recursion of Kennett [27], computed
537 independently, to 10^{-14} . The schemes are those of §§7–8, with the contrast
538 now changing from plane to plane; nothing else changes. The errors are in
539 Table 6 and Fig. 5(c).

540 *The interfaces cost nothing.* Every order is the one measured for the uniform
541 layer. At a model interface the strain of the true field jumps. A voxel basis
542 that is discontinuous from voxel to voxel represents that jump exactly when
543 the interface lies on a voxel boundary, so the interface adds no error of its
544 own. The only requirement is that voxel boundaries coincide with the model's
545 interfaces. A voxel that straddled an interface would carry a contrast that is
546 not uniform within it, a case not examined here.

Table 6: Relative error against the exact stratified layer, for m planes of voxels per model layer (§9). At normal incidence the order is measured from $m = 4$ to 16, where it is also 2.00 for collocation and 4.00 in transmission; at 20° it is measured from $m = 2$ to 4.

		$m = 1$	$m = 2$	$m = 4$	order
P, normal, R	mean only	1.9×10^{-3}	4.7×10^{-4}	1.2×10^{-4}	2.00
	first moment	8.5×10^{-7}	5.3×10^{-8}	3.3×10^{-9}	4.00
S, normal, R	mean only	5.3×10^{-3}	1.3×10^{-3}	3.3×10^{-4}	2.00
	first moment	6.2×10^{-6}	3.9×10^{-7}	2.4×10^{-8}	4.00
P, 20° , R_{PP}	mean only	1.4×10^{-3}	3.6×10^{-4}	9.0×10^{-5}	2.00
	first moments	3.3×10^{-7}	2.1×10^{-8}	1.3×10^{-9}	4.00
P, 20° , R_{PS}	mean only	2.8×10^{-3}	6.9×10^{-4}	1.7×10^{-4}	2.00
	first moments	1.4×10^{-6}	8.8×10^{-8}	5.5×10^{-9}	4.00

547 *What fourth order buys.* For a reflection error of 10^{-6} at normal incidence
 548 (P), the mean-only voxel needs 347 planes of voxels, about 43 per model
 549 layer. The first-moment voxel needs 8, one per model layer: the model's own
 550 layering, which already gives 8.5×10^{-7} . Counting unknowns (9 per voxel
 551 against 36), that is 3123 against 288 per vertical column of voxels, a factor
 552 of eleven. In three dimensions, with the lateral grid refined as the vertical
 553 one is, the number of voxels falls by $(347/8)^3 \approx 8 \times 10^4$ and the number of
 554 unknowns by about 2×10^4 . The same model has been solved independently
 555 in Python, with a contrast for each plane: the raw reflected displacements
 556 agree with the Mathematica scheme to 1.1×10^{-15} at normal incidence and
 557 at 20° , with and without the first moments. Lateral heterogeneity needs the
 558 full three-dimensional solver; it is the first extension discussed below.

559 10. A sphere: what the layer predicts in three dimensions

560 The layer and the stratified layer are laterally uniform. A sphere is not,
561 and it has an exact solution for elastic scattering [28, 29]: it is the usual
562 benchmark for volume-integral and discrete-dipole methods. This section
563 solves it with the full three-dimensional voxel scheme, to test what the layer
564 predicts for a laterally heterogeneous body. The sphere is voxelised by the
565 cubes of an n_{sub}^3 grid whose centres lie inside it, and the Foldy–Lax system
566 is solved with the uniform-field voxel (the second-order scheme of §§6–7;
567 the first-moment voxel of §8 has not been built for finite three-dimensional
568 scatterers) and the cell-averaged coupling, by an FFT-accelerated iterative
569 solver. The single site is the package’s cube T -matrix, which differs from the
570 collocation closure only through its far-field corrections (§6.4) and has the
571 same order. A P wave is incident; the contrast is that of §7. The error is the
572 largest difference between the scattered far field and the exact one over nine
573 scattering angles from 11° to 169° , relative to the largest exact value, at a
574 distance of 5×10^4 radii where the far field has converged (`scripts/pilot_`
575 `sphere_voxel_vs_mie.py`).

576 *The raw error does not converge.* Table 7 shows the error for grids of 4, 6 and
577 8 voxels across. It does not fall steadily: at $k_S a = 0.5$ it is 6.1%, then 17%,
578 then 2.4%. The staircase changes with every grid, and with it the volume and
579 the shape of the scatterer. Rescaling by the ratio of the sphere’s volume to
580 the staircase’s removes the largest swings but leaves an error of about 2% at
581 $k_S a = 0.5$, and 8 to 11% at $k_S a = 1.0$, that does not decrease. A refinement
582 study of the voxelised sphere measures the staircase, not the scheme.

Table 7: A sphere voxelised on grids of n_{sub} voxels across, against exact elastic scattering: relative far-field error, raw and rescaled by the ratio of the sphere’s volume to the staircase’s. It does not converge with refinement, because the staircase changes with the grid.

n_{sub}	voxels	$k_S a = 0.5$		$k_S a = 1.0$	
		raw	volume-rescaled	raw	volume-rescaled
4	32	6.1×10^{-2}	1.8×10^{-2}	1.19×10^{-1}	7.7×10^{-2}
6	136	1.71×10^{-1}	2.8×10^{-2}	6.8×10^{-2}	1.12×10^{-1}
8	280	2.4×10^{-2}	2.1×10^{-2}	4.6×10^{-2}	8.6×10^{-2}

583 *Refining a fixed shape.* The staircase is itself a union of cubes, so it has its
584 own exact solution, and the voxel scheme converges to it when each voxel
585 is split into m^3 sub-voxels of the same shape (Fig. 4). The change of the
586 far field under that refinement is the discretisation error alone; the differ-
587 ence between the converged staircase field and the sphere’s is the shape error
588 alone. Table 8 refines the $n_{\text{sub}} = 8$ staircase to $m = 2$ and 3, up to 7560
589 voxels (`scripts/pilot_sphere_staircase_refinement.py`). The discreti-
590 sation error converges at order 1.99 at $k_S a = 0.5$ and 2.06 at $k_S a = 1.0$
591 (Fig. 5d): the second order of the uniform-field voxel on the layer, now on
592 a laterally heterogeneous body. At the original grid it is 5.4×10^{-4} and
593 2.7×10^{-3} , larger by a factor of about five when the frequency doubles, close
594 to the factor four of its $(kh)^2$ scaling. The shape error, from the field ex-
595 trapolated to $m \rightarrow \infty$, is 2.36×10^{-2} and 4.83×10^{-2} : 44 and 18 times the
596 discretisation error. On a sphere the staircase dominates the error of the
597 scheme, which is why the layer, where the staircase is exact, is the test that
598 measures the scheme itself.

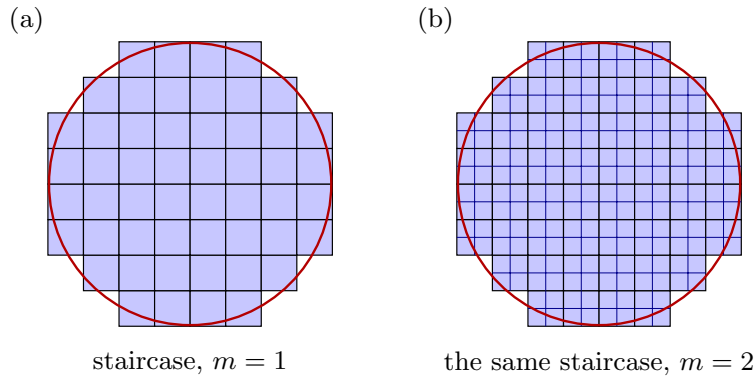


Figure 4: Refining a fixed shape. (a) A cross-section through the $n_{\text{sub}} = 8$ voxelisation of a sphere (red circle): the voxels whose centres lie inside it. (b) The same staircase with every voxel split into $2 \times 2 \times 2$ sub-voxels ($m = 2$): the shape, and so its exact solution, is unchanged, and only the discretisation is refined.

Alt text: Two panels, each a square grid of shaded cells approximating a circle drawn in red; the shaded cells form a stepped outline that crosses the circle. In the right panel every shaded cell is divided into four smaller cells by thin lines, while the stepped outline is exactly the same as on the left.

Table 8: The $n_{\text{sub}} = 8$ staircase with every voxel split into m^3 sub-voxels: far-field error against the exact sphere, and the change from the previous level. The change is the discretisation error; its order is measured from the two changes, and the field extrapolated to $m \rightarrow \infty$ gives the shape error.

m	voxels	$k_{\text{sa}} = 0.5$		$k_{\text{sa}} = 1.0$	
		vs sphere	change	vs sphere	change
1	280	2.403×10^{-2}		4.604×10^{-2}	
2	2240	2.369×10^{-2}	4.04×10^{-4}	4.779×10^{-2}	2.05×10^{-3}
3	7560	2.364×10^{-2}	7.51×10^{-5}	4.811×10^{-2}	3.65×10^{-4}
order of the discretisation error		1.99		2.06	
shape error ($m \rightarrow \infty$)		2.36×10^{-2}		4.83×10^{-2}	

599 11. Discussion

600 Figure 5 collects the convergence of every test, with the order fitted to
 601 each by least squares (`scripts/plot_convergence_orders.py`).

602 *Where the error of a voxel scheme lives.* The results separate the two ap-
 603 proximations a voxel scheme makes. The *coupling* between voxels makes
 604 none: averaged over the source cell it is the continuum’s cell integral ex-
 605 actly, between planes by the form factor’s zeros and within a plane by the
 606 tiling identity, and the self term it omits is exactly the one the single site
 607 carries. Every error therefore comes from the *single site*, from the internal
 608 field it assumes. For a uniform field that error is the midpoint rule’s, known
 609 in closed form; for a linear field it is the next term of the same expansion.
 610 The converse is the practical warning. A kernel that is not the cell integral
 611 (the point Green’s tensor, or a cell average that does not match the single
 612 site’s closure) leaves a local term that does not vanish as the voxels shrink
 613 (§4): the evanescent lattice orders carry a static contribution of fixed size per
 614 voxel volume. A scheme built that way converges to the wrong limit, and
 615 refinement cannot reveal it, because the error is scale-invariant.

616 *What the single site is.* The moment hierarchy of §6 turns the uniform-field
 617 single site from a modelling choice into a derived object with a stated error.
 618 Its static part is exact for a sphere, and for a cube it differs from the sphere’s
 619 only by the cubic split of Eq. (24). Its dynamic part departs from exact
 620 scattering by a sphere only through the real $(ka)^2$ term of Eq. (25); the
 621 radiation damping is exact. That term and the collocation error of Eq. (28)
 622 have the same origin, the variation of the field across the scatterer, and the

623 same remedy, a richer internal field. In a lattice the remedy is cheap: the
 624 field a voxel must represent is the smooth continuum field, not the field of
 625 an isolated cube, with its concentrations at the edges and corners.

626 *Relation to the discrete dipole approximation.* The same partition appears in
 627 the electromagnetic literature as the choice of polarisability. The Clausius–
 628 Mossotti relation, the lattice dispersion relation [6] and the integrated Green
 629 tensor [7] all adjust the single site so that the lattice reproduces the con-
 630 tinuum. For space-filling cubes the adjustment is neither needed nor a free
 631 parameter: the tiling identity fixes the local field exactly, the self term of
 632 the collocation closure is the one that makes it cancel, and what is left is a
 633 closed-form error rather than an empirical one. The second-order convergence
 634 reported for the DDA by Yurkin et al. [1] corresponds to the uniform-field
 635 voxel’s, and the opposite-signed constants of Eq. (28) explain why a polaris-
 636 ability correction can improve one observable only at the expense of another,
 637 as the package’s own form-factor corrections do in reflection and transmis-
 638 sion. Going beyond second order requires a richer internal field, which is the
 639 point of higher-order bases for volume integral equations; the first-moment
 640 voxel is the lowest such basis, and in its spectral form the Galerkin double
 641 average makes the lattice sum converge without Ewald splitting.

642 *The cost of fourth order.* A first-moment voxel carries four times as many
 643 unknowns as a uniform one (36 against 9 in three dimensions). The closed
 644 forms show what that buys. To reach an error ε in reflection, the uniform
 645 voxel needs $kd = \sqrt{8\varepsilon}$ and the first-moment voxel $kd = (720\varepsilon)^{1/4}$; at $\varepsilon = 10^{-6}$
 646 that is $kd = 2.8 \times 10^{-3}$ against 0.16, voxels 58 times larger in each direction.
 647 In a three-dimensional model the voxel count falls by $58^3 \approx 2 \times 10^5$ and the

648 number of unknowns by about 5×10^4 (these are the Born-order constants;
 649 at the contrast studied the first-moment voxel's measured errors lie within
 650 5% of them, while the uniform voxel's transmission constant grows with the
 651 contrast, so the comparison is if anything conservative). The stratified model
 652 of §9 measures it directly: 347 planes of uniform-field voxels against 8 first-
 653 moment voxels for the same accuracy. Higher moments continue the trend
 654 in one dimension: by Eq. (32) the second-moment voxel reaches $\varepsilon = 10^{-6}$ at
 655 $kd = 0.68$ and the third-moment voxel at $kd = 1.5$, where a voxel is already
 656 a quarter of a wavelength. A three-dimensional voxel of degree p has not
 657 been built; with the Legendre polynomials of total degree at most p it would
 658 carry 10 ($p = 2$) or 20 ($p = 3$) weights per state component, against 4 for
 659 the first-moment voxel. The gain is largest precisely where it matters: when
 660 the model varies on scales that are coarse compared with the accuracy the
 661 wavefield must have.

662 *Limitations and outlook.* The results concern laterally uniform media, a ho-
 663 mogeneous layer and a randomly stratified one, for which the discrete system
 664 is exactly a chain of planes; a laterally heterogeneous medium needs the first-
 665 moment voxel in a full three-dimensional solver, where the gradient channels
 666 couple through the evanescent lattice orders (§8) and the tiling identity holds
 667 only to $O(k_{\parallel}h)$. The sphere of §10 shows that the second order of the uniform-
 668 field voxel carries over to a laterally heterogeneous body; the fourth order
 669 has still to be shown there. The numerical evidence is at moderate, physi-
 670 cally typical contrasts (up to 15% in density and moduli) and at $k_S h \leq 0.3$,
 671 the range in which the single-site T -matrix is validated; the closed forms are
 672 first order in the contrast, and the departures from them at the full contrast

673 are of second order. What is established is the *order* of convergence and
 674 its leading constant, not the behaviour at strong contrast, where the single
 675 site's own closure may limit the accuracy before the lattice coupling does.
 676 The single site's dynamic error is in closed form for a sphere only; for a cube,
 677 for which no exact solution exists, it is known through the layer (§§7–8).
 678 Two extensions follow directly. The first is the three-dimensional solver it-
 679 self. The second is a stratified *background*, a layered reference medium, as
 680 distinct from the stratified contrast of §9. There the lattice coupling then
 681 carries the reverberation of the layering, which is not translation invariant in
 682 depth, and whether a tiling identity holds for it is the natural next question.

683 12. Conclusions

684 A lattice of cubic voxels reproduces a continuous elastic medium in a
 685 precise sense. Because cubes tile the plane, the source-cell-averaged cou-
 686 pling between voxels is the continuum's cell integral exactly, and at normal
 687 incidence the lattice kernel is available in closed form with no lattice sum.
 688 The single site is derived rather than assumed. The moments of the Green's
 689 tensor over the cube, taken as distributions, give a closed hierarchy whose
 690 first-gradient truncation is the uniform-strain closure. That closure is exact
 691 for a sphere in the static limit, and it departs from exact scattering by a
 692 sphere only by a real $(ka)^2$ term, with the radiation damping exact. For
 693 a cube it differs from the sphere only by a cubic anisotropy of the shear
 694 response. The collocation closure carries the same self term as the lattice
 695 kernel, which cancels, so the discrete system is a collocation of the contin-
 696 uum equation. It converges at second order, with a leading error that is a

midpoint-rule error of the internal field, known in closed form for every discretisation, and opposite in sign in reflection and transmission. A voxel that also carries the first moment of its internal field converges at fourth order, at normal incidence (where the three-dimensional scheme reduces exactly to one dimension) and at oblique incidence, with a leading error, $-(kd)^4/720$, that no longer depends on the direction of observation. With Legendre moments to degree p it converges at order $2p + 2$ with the closed-form constant of Eq. (32), and a single plane of such voxels reproduces the thin-layer series term by term through $O(D^{2p+2})$. The orders survive the interfaces of a randomly stratified model, provided voxel boundaries coincide with its interfaces, and there the first-moment voxel reaches an accuracy of 10^{-6} with one voxel per model layer, where the uniform-field voxel needs about forty. On a voxelised sphere the discretisation error is again second order, and the staircase approximation of the surface dominates it, by factors of 18 to 44. The results, and the evidence for each, are collected in Table 9.

Acknowledgments

This work began thirty years ago in the author’s doctoral research at the Research School of Earth Sciences of the Australian National University, Canberra [9].

Declaration of competing interest

The author declares no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Table 9: The results of the paper, and the evidence for each.

Result	Evidence
Continuous layer, exact and to $O(D^k)$	reference to 2.5×10^{-8} ; series order k
Thin-layer series, Eq. (5)	each coefficient symbolically, to D^4
Point-kernel local term survives $d \rightarrow 0$	$11 \times$ the continuum term at $m = 1$
Averaged coupling between planes = continuum $\times$ sinc	2.8×10^{-16} , 81 entries
Tiling identity, Eq. (11)	21/21 components to 4×10^{-9}
Exact kernel, Eq. (12)	replaces a 9.4×10^{-4} static error
Moments as distributions, Eq. (19)	sum rule (20) exact; M, Q equal an independent archive, 81/81
Channels (21)–(23)	a production code by another route, 2×10^{-15}
Cube vs sphere, Eq. (24)	symbolic; and numerically at five ν , 4×10^{-16}
Ball closure = exact Mie, static	2×10^{-10} ; cube's bulk and shear average 10^{-8}
Ball closure vs Mie, dynamic, Eq. (25)	symbolic; numerical Mie 5×10^{-4} ; $(ka)^3$ terms exact
Self term cancels, Eq. (27)	4×10^{-16}
Package T = closure statically	2×10^{-5} at $k_S h = 0.01$
Second-order convergence	order 2.000, three frequencies
Closed-form error, Eq. (28)	$n = 1$ –16, to 10^{-4} of the error
First-moment voxel, Eq. (30)	order 4.00, three frequencies
First-moment voxel, oblique, Eq. (33)	order 4, P and S, 20° and 40°

720 **Data availability**

721 The Mathematica notebooks and Python scripts that derive and check ev-
722 ery result, and the package they test, are archived at [https://doi.org/10.](https://doi.org/10.5281/zenodo.23003241)
723 5281/zenodo.23003241 and developed at [https://github.com/tmnestor/](https://github.com/tmnestor/MultipleScatteringCalculations)
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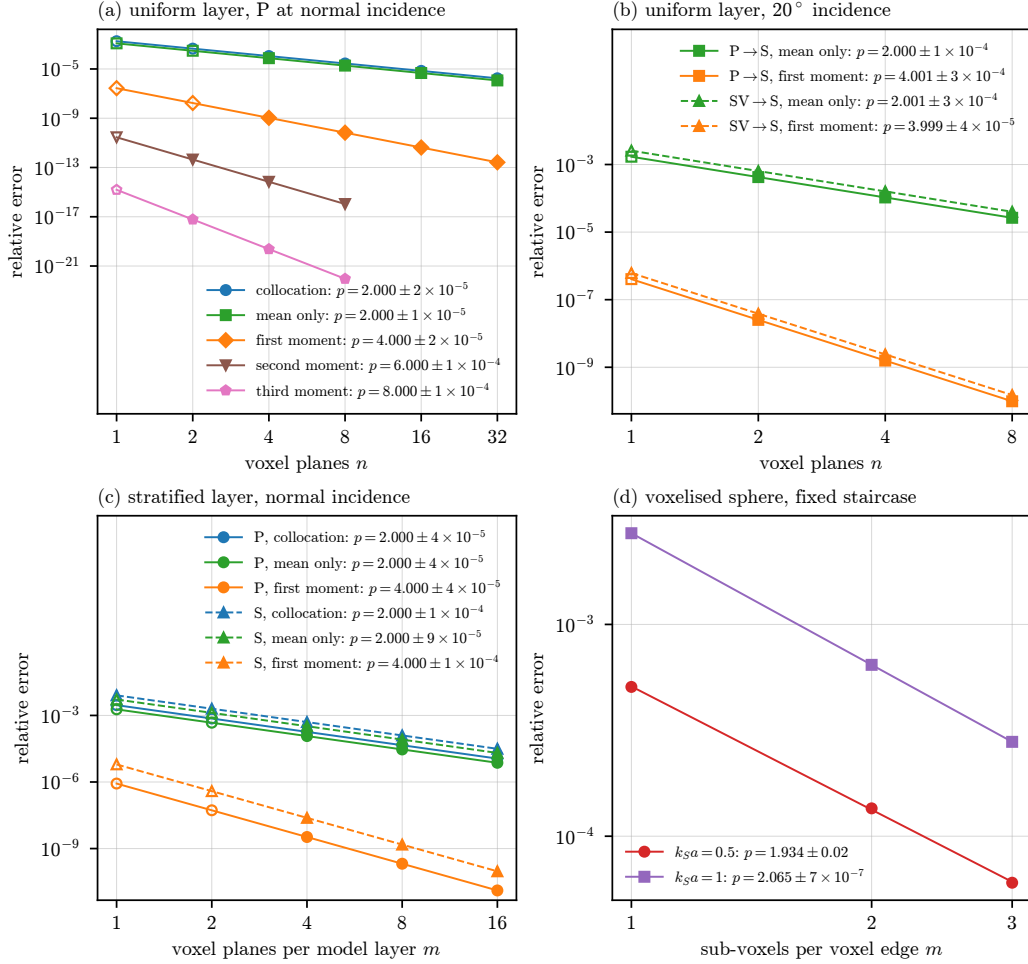


Figure 5: Convergence of every test, with least-squares regression lines. Each line fits $\log e = c - p \log n$ to the filled points; open points are pre-asymptotic and are not fitted. The legend gives the order p and its standard error. Colour marks the scheme: collocation (§6), the mean-only voxel and the voxels carrying Legendre moments to degree $p = 1, 2$ and 3 (Eqs. (30) and (33)). (a) The uniform layer, P reflection at normal incidence ($D = 2$ m, $\omega = 300$ rad s $^{-1}$, the contrast of §7), fitted from $n = 4$; the second- and third-moment voxels, computed in 40-digit arithmetic, fitted from $n = 2$. (b) The uniform layer at 20°, fitted from $n = 2$: the S wave converted from incident P (solid) and reflected from incident SV (dashed). (c) The stratified layer of §9, reflection at normal incidence for P (solid) and S (dashed), against the number m of voxel planes per model layer, fitted from $m = 4$. (d) The voxelised sphere of §10: the far-field difference between the $n_{\text{sub}} = 8$ staircase with every voxel split into m^3 sub-voxels and the staircase field extrapolated to $m \rightarrow \infty$, fitted over $m = 1, 2, 3$. That limit is extrapolated from the same three levels, so panel (d) checks consistency; its fitted orders differ slightly from those of Table 8 (1.99 and 2.06), which are measured from the changes between levels.

Alt text: Four log-log panels of relative error against refinement, each showing data